

Fighting disease

Your body has powerful defenses to fight off the germs that cause disease, but sometimes you might need extra help from doctors to get well. Doctors use a range of techniques to determine what kind of illness you might have and how best to treat it.

IDENTIFYING GERMS

The doctor's first job is to make a diagnosis, which means determining what kind of disease you have. If the doctor thinks the disease is caused by germs, you may need to provide a sample. This will be sent to a laboratory, which will identify any germs present.



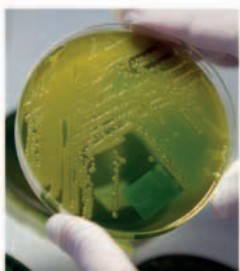
◀ TAKING SAMPLES

While examining a young girl, a doctor takes a sample of saliva from her mouth with a cotton swab. The sample is then sent to a lab.



◀ SAMPLE PLATE

In the lab, the sample is smeared across a plate of jelly. The jelly contains nutrients that encourage bacteria from the sample to grow and multiply.



◀ CULTURE

Over several days, each bacterium from the sample multiplies to form a colony of thousands, forming a visible spot on the sample plate.



◀ IDENTIFY

The bacteria can now be identified. Viewing them with a microscope reveals their distinctive size, shape, and color.

HYGIENE

Dirty hands can spread disease from person to person. This picture shows colonies of bacteria that were allowed to grow from a dirty handprint. The bacteria are *E. coli*, which can cause food poisoning if they get into food. Good hygiene can help prevent the spread of disease. For example, washing your hands after using the bathroom reduces the risk of passing harmful bacteria to others.

GERM KILLERS

Diseases caused by bacteria are treated with drugs called antibiotics. These attack and destroy bacterial cells without harming body cells. There are many kinds of antibiotics, each active against particular kinds of bacteria. For diseases caused by viruses, doctors may prescribe antiviral drugs instead.

